have elaborate requirements of specific nutrients like vitamins and growth promoting subtances and or not easily pleased or satisfied by ordinary nutrients available in nature. Example: Blood agar is used to identify haemolytic bacteria (Figure 5.4) and Chocolate agar used to identify *Neisseria gonorrhoeae.*



Figure 5.4: Blood Agar showing alpha, beta & gamma – haemolytic colonies



In 1919, James Brown used blood agar as diagnostic medium to study the haemolytic patterns of bacteria.

iii) Selective medium

Selective medium contains one or more agents (selective components) that inhibit unwanted organisms but allow the desired organisms to grow. Growth of unwanted microbes is suppressed by adding bile salts, antibiotics and dyes. Example: Mannitol salt agar is selective for *Staphylococci.* This medium contains 7% Sodium chloride that inhibits the growth of other bacterial population but allows the growth of *Staphylococci* (Figure 5.5). Moreover it has Phenol red dye to indicate acid production. *Staphylococcus* utilizes Mannitol and produces acid which changes the colour of the Phenol red indicator to yellow. Salmonella-Shigella (SS) agar is selective for *Salmonella* (Figure 5.6).



Figure 5.5: Growth of *Staphylococcus aureus* on Mannitol salt agar

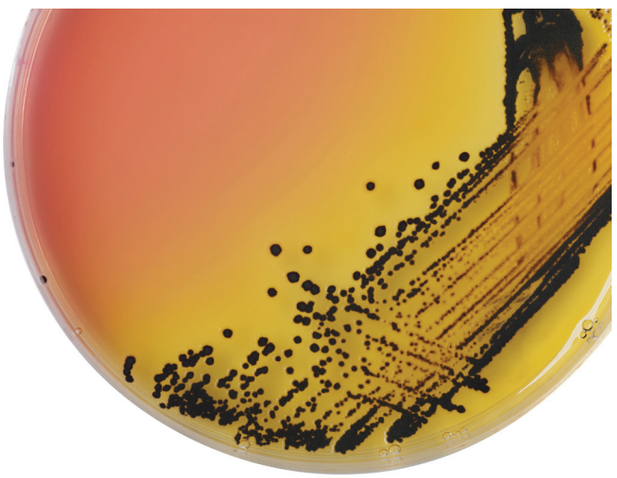


Figure 5.6: Growth of *Salmonella* on SS agar

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